

# Parthiv Jasoliya

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## Education

| Course | Institute                                           | Year        | CPI/% |
|--------|-----------------------------------------------------|-------------|-------|
| B.Tech | Sardar Vallabhbhai National Institute of Technology | 2022 – 2026 | 8.16  |

## Experience

- TruEstate – Software Engineering Intern** | *Next.js, NestJS, TypeScript, GCP, Qdrant* **Jan 2026 – Present**
- Designed and deployed scalable cloud-native systems using GCP (Cloud Run, App Engine, Compute Engine, Cloud SQL, BigQuery, Cloud Scheduler, Pub/Sub, Cloud Tasks) for Canvas Homes.
  - Launched a high-performance property matching engine using vector embeddings and Qdrant, computing similarity scores across all listings to deliver intelligent match percentages for buyers.
  - Architected an AI-powered email communication platform (custom-built email client replica) with prediction models and semantic analysis by integrating generative AI APIs, improving sales team efficiency.
- TCS R&I – Software Engineering Intern** | *Python, AWS, LLMs, Agentic Frameworks* **May 2025 – Jul 2025**
- Designed a hybrid ingestion pipeline utilizing Python scripts for high-volume structured feeds (STIX/JSON) and a fine-tuned LLM to extract entities from unstructured vendor reports, processing 100,000+ events/day.
  - Automated CIS benchmark checks across 50+ AWS resources using Python Lambda agents, identifying and mitigating 120+ security misconfigurations pre-deployment.
  - Evaluated 10,000+ IAM access patterns using custom AI agents, leading to a 30% reduction in overly permissive policies.
- Voxel Verse – Software Engineering Intern** | *Python, Diffusion Models, GANs* **Jul 2024 – Aug 2024**
- Architected a Stable Diffusion and PyTorch pipeline to generate 500+ 2D game assets per hour, seamlessly integrating colliders and rigid-body parameters into the game engine.
  - Optimized the generation workflow by batching denoising steps and improving GPU utilization, reducing latency from 6.0s to 5.1s (15%) across a 1,000-asset benchmark while maintaining consistent visual fidelity.
  - Enhanced environment pattern generation, increasing designer-approval rate from 72% to 79% through A/B testing.

## Projects

- Strata – Quant Trading Platform** | *Python, C++, Polars, React* **Aug 2024**
- Developed a high-performance C++ Monte Carlo engine exposed to Python via PyBind11, simulating 10,000+ price paths for VaR and CVaR risk estimation.
  - Implemented Hierarchical Risk Parity using unsupervised clustering to construct stable portfolios, reducing out-of-sample variance by 15% compared to traditional optimization across a 5-year backtest.
  - Formulated a statistical arbitrage service using Engle–Granger cointegration to identify mean-reverting pairs from 500+ equities, achieving a simulated Sharpe ratio of 1.85.
- Graph Visualizer – Algorithm Analysis Tool** | *React, TypeScript, Tailwind CSS* **Dec 2023**
- Built a graph editor supporting multiple input formats (edge lists, adjacency lists, parent–child arrays).
  - Implemented graph analysis features including components/SCC, MST, bridges & articulation points, and tree/bipartite visualizations and garnered over 60 users.
  - Optimized force-layout rendering to smoothly visualize graphs with ~1000 nodes and added interactive modes (lock, color, multi-testcases).

## Technical Skills

**Languages:** C++, Python, Java, TypeScript, JavaScript, HTML5, CSS3.  
**Backend:** RESTful APIs, FastAPI, Node.js, Express, Spring Boot, NestJS.  
**Frontend:** React, Next.js.  
**Cloud & DevOps:** GCP, AWS, Docker, Git, GitHub, GitLab, Postman.  
**Databases:** MySQL, MongoDB, Firebase, Qdrant.  
**AI / Tools:** Stable Diffusion, LLMs, Agentic Frameworks, Polars.

## Certifications & Achievements

- Codeforces:** Max Rating 2338 (**International Master**) – 800+ Problems Solved.
- LeetCode:** Max Rating 2409 (**Guardian**) – 600+ Problems Solved.
- CodeChef:** Max Rating 2102 (**5 Star**) – 200+ Problems Solved.
- Ranked **#130/22,000+** in **Meta Hacker Cup 2025 R3** and **#581** in 2024 R1.
- ICPC Regionalist** (Amritapuri Regional Round) – 2024 and 2025.
- Hackathon Finalist** – Precision Healthcare Challenge 2025 (GE Healthcare).
- Amazon ML Summer School 2024** – Selected among **3,000** participants.